

$$\begin{aligned} &\Rightarrow 4(b^4 + a^2c^2 - 2ab^2c) \\ &\quad - 4(a^2c^2 - bc^3 - a^3b + ab^2c) = 0 \\ &\Rightarrow 4b^4 + 4a^2c^2 - 8ab^2c - 4a^2c^2 \\ &\quad + 4bc^3 + 4a^3b - 4ab^2c = 0 \\ &\Rightarrow 4b^4 - 12ab^2c + 4bc^3 + 4a^3b = 0 \\ &\Rightarrow b^3 + c^3 + a^3 - 3abc = 0 \\ &\therefore a^3 + b^3 + c^3 = 3abc \end{aligned}$$

101. (a) Given m and n are the roots of the given equation $ax^2 + bx + c = 0$

$$\therefore \text{Sum of roots} = m + n = -b/a \quad \dots(i)$$

$$\text{and product of roots} = mn = \frac{c}{a} \quad \dots(ii)$$

$$\begin{aligned} \text{Now, } \frac{m^2 + 1}{m} + \frac{n^2 + 1}{n} &= \frac{m^2n + n + mn^2 + m}{mn} \\ &= \frac{mn(m + n) + (m + n)}{mn} \\ &= \frac{(m + n)(mn + 1)}{mn} \\ &= \frac{(m + n)(\frac{c}{a} + 1)}{\frac{c}{a}} = \frac{-b(a + c)}{ac} \end{aligned}$$

[using Eqs. (i) and (ii)]

$$\begin{aligned} \text{and } \frac{m^2 + 1}{m} \times \frac{n^2 + 1}{n} &= \frac{(m^2 + 1)(n^2 + 1)}{mn} \\ &= \frac{m^2n^2 + n^2 + m^2 + 1}{mn} \\ &= \frac{(mn)^2 + (m + n)^2 - 2mn + 1}{mn} \\ &= \frac{\left(\frac{c}{a}\right)^2 + \left(\frac{-b}{a}\right)^2 - 2\left(\frac{c}{a}\right) + 1}{\frac{c}{a}} \\ &= \frac{c^2 + b^2 - 2ac + a^2}{ac} \\ &= \frac{b^2 + (a - c)^2}{ac} \end{aligned}$$

We know that, quadratic equation can be written as

$$x^2 - (\text{Sum of roots})x + \text{Product of roots} = 0$$

$$\Rightarrow x^2 - \left(\frac{-b(a + c)}{ac}\right)x + \left(\frac{b^2 + (a - c)^2}{ac}\right) = 0$$

$$\begin{aligned} &\Rightarrow acx^2 + b(a + c)x + b^2 + (a - c)^2 = 0 \\ &\Rightarrow acx^2 + (ab + bc)x + b^2 + (a - c)^2 = 0 \end{aligned}$$

102. (c) When mistake is done in first degree term, then the roots of the equation are -9 and -1 .

$$\therefore \text{Equation is } (x + 1)(x + 9) = x^2 + 10x + 9 \quad \dots(i)$$

When mistake is done in constant term, then the roots of equation are 8 and 2 .

$$\therefore \text{Equation is } (x - 2)(x - 8) = x^2 - 10x + 16 \quad \dots(ii)$$

From Eqs. (i) and (ii), we get

$$x^2 - 10x + 9 = 0 \text{ is the correct equation.}$$

103. (c) Given equation,

$$\begin{aligned} 7(x + 2a)^2 + 3a^2 &= 5a(7x + 23a) \\ \Rightarrow 7(x^2 + 4a^2 + 4ax) + 3a^2 &= 35ax + 115a^2 \\ \Rightarrow 7x^2 - 7ax - 84a^2 &= 0 \\ \Rightarrow x^2 - ax - 12a^2 &= 0 \\ \Rightarrow (x + 3a)(x - 4a) &= 0 \\ \Rightarrow x = -3a \text{ and } x = 4a \end{aligned}$$

Since, m and n are the roots of the given equation.

$$\text{Let } m = 4a \text{ and } n = -3a (\because m > n)$$

$$\begin{aligned} \therefore 3m - n &= 3(4a) - (-3a) \\ &= 12a + 3a = 15a \end{aligned}$$

104. (a) Given equation is $px^2 + qx + r = 0$.

Let one root of the equation be α .

Then, other root $= 3\alpha$

$$\begin{aligned} \therefore \text{Sum of roots} &= \alpha + 3\alpha = -\frac{q}{p} \\ \Rightarrow 4\alpha &= -\frac{q}{p} \Rightarrow \alpha = -\frac{q}{4p} \quad \dots(i) \end{aligned}$$

$$\text{and product of roots} = (\alpha) \cdot (3\alpha) = \frac{r}{p}$$

$$\begin{aligned} \Rightarrow 3\alpha^2 &= \frac{r}{p} \Rightarrow 3\left(-\frac{q}{4p}\right)^2 = \frac{r}{p} \\ &\quad [\text{from Eq. (i)}] \end{aligned}$$

$$\Rightarrow \frac{3q^2}{16p^2} = \frac{r}{p} \Rightarrow 3q^2p = 16p^2r$$

$$\Rightarrow 3q^2 = 16pr$$

105. (b) I. Given, m and n are the roots of the equation

$$x^2 + ax + b = 0.$$

$$\therefore \text{Sum of roots, } m + n = -a \quad \dots(i)$$

$$\text{and product of roots, } mn = b \quad \dots(ii)$$

Also, given m^2 and n^2 are the roots of the equation

$$x^2 - cx + d = 0.$$

$$\therefore m^2 + n^2 = c \quad \dots(iii)$$

$$\text{and } m^2n^2 = d \quad \dots(iv)$$

On squaring both sides of Eq. (i), we get

$$m^2 + n^2 + 2mn = a^2$$

$$[\text{from Eqs. (i) and (ii)}]$$

$$\Rightarrow c + 2b = a^2 \Rightarrow c = a^2 - 2b$$

$$\Rightarrow 2b - a^2 = -c$$

Hence, statement I is not correct.

II. From Eq. (ii), $m^2n^2 = b^2$

$$\Rightarrow b^2 = d$$

Hence, statement II is correct.

106. (a) If a is positive and $b^2 - 4ac \leq 0$, then the sign of quadratic polynomial $ax^2 + bx + c$ is always positive.

107. (c) We have, $x + \frac{100}{x} > 50$

$$\text{and } 1 \leq x \leq 100 \Rightarrow x^2 - 50x + 100 > 0$$

$$\begin{aligned} \Rightarrow x &= \frac{50 \pm \sqrt{2500 - 400}}{2} \\ &= \frac{50 \pm 10\sqrt{21}}{2} \end{aligned}$$

$$x < 25 - 5\sqrt{21} \text{ or } x > 25 + 5\sqrt{21}$$

$$\Rightarrow x < 2.087 \text{ or } x > 47.91$$

$$\Rightarrow x = 1 \text{ and } 2 \text{ or } x = 48, 49, \dots, 100$$

$\therefore$ Number of total values of

$$x = 2 + 53 = 55$$

108. (c) We have, $x^2 - 4x - \log_{10} N = 0$

We know that, roots are real, if $D \geq 0$

$$\Rightarrow 16 - 4(-\log_{10} N) \geq 0$$

$$\Rightarrow 16 + 4 \log_{10} N \geq 0$$

$$\Rightarrow \log_{10} N \geq -4$$

$$\Rightarrow N \geq 10^{-4} \Rightarrow N \geq \frac{1}{10000}$$

$$\text{Minimum value of } N = \frac{1}{10000}$$

109. (c) Let $x = \sqrt{4 + \sqrt{4 - \sqrt{4 + \dots}}}$

$$\therefore x = \sqrt{4 + \sqrt{4 - x}}$$

$$\Rightarrow x^2 = 4 + \sqrt{4 - x}$$

$$\Rightarrow (x^2 - 4)^2 = 4 - x$$

$$\Rightarrow (x^2 - 4)^2 + x = 4 \quad \dots(i)$$

$$\text{For option (c), } x = \frac{\sqrt{13} + 1}{2}$$

From Eq. (i),

$$\begin{aligned} \text{LHS} &= \left[\left(\frac{\sqrt{13} + 1}{2} \right)^2 - 4 \right]^2 + \frac{\sqrt{13} + 1}{2} \\ &= \left[\frac{13 + 1 + 2\sqrt{13} - 16}{4} \right]^2 + \frac{\sqrt{13} + 1}{2} \\ &= 4 = \text{RHS} \end{aligned}$$

110. (c) We have, $-2 \leq x \leq 1 \quad \dots(i)$

$$-1 \leq y \leq 2$$

$$\therefore -2 \leq -y \leq 1 \quad \dots(ii)$$

$$\text{and } 3 \leq z \leq 6$$

$$6 \leq 2z \leq 12 \quad \dots(iii)$$

On adding Eqs. (i), (ii) and (iii), we get

$$-2 - 2 + 6 \leq x - y + 2z \leq 1 + 1 + 12$$

$$2 \leq k \leq 14$$

- 111. (a)** Let α and β be the roots of equation $ax^2 + bx + c = 0$.

$$\text{Then, } \alpha + \beta = \frac{-b}{a}$$

$$\text{and } \alpha\beta = \frac{c}{a}$$

$$\text{Now, } \alpha + \beta = \left(\frac{1}{\alpha}\right)^2 + \left(\frac{1}{\beta}\right)^2$$

$$\Rightarrow \alpha + \beta = \frac{\beta^2 + \alpha^2}{(\alpha\beta)^2} = \frac{(\alpha + \beta)^2 - 2\alpha\beta}{(\alpha\beta)^2}$$

$$\Rightarrow \frac{-b}{a} = \frac{b^2/a^2 - 2c/a}{c^2/a^2}$$

$$\Rightarrow ab^2 - 2a^2c = -bc^2$$

$$\Rightarrow ab^2 + bc^2 = 2a^2c$$

- 112. (d)** Here, $a = 1, b = -p, c = q$

$$\text{and } a' = 1, b' = q, c' = -p$$

Applying common root condition,

$$(a'c - ac')^2 = (bc' - b'c)(ab' - a'b)$$

$$\Rightarrow (q + p)^2 = (p^2 - q^2)(q + p)$$

$$\Rightarrow (p + q) = (p - q)(p + q)$$

$$\Rightarrow p - q = 1$$

$$\Rightarrow p - q - 1 = 0$$

- 113. (c)** Let α and α^2 be the roots of equation $x^2 + px + q = 0$.

$$\text{Then, } \alpha \cdot \alpha^2 = q$$

$$\therefore \alpha = (q)^{1/3} \quad \dots(i)$$

$$\text{and } \alpha + \alpha^2 = -p$$

$$\therefore (q)^{1/3} + (q)^{2/3} = -p \quad \dots(ii)$$

On cubing both sides, we get

$$q + q^2 + 3q(q^{1/3} + q^{2/3}) = -p^3$$

$$\Rightarrow q + q^2 + 3q(-p) = -p^3$$

[from Eq. (ii)]

$$\Rightarrow q + q^2 + p^3 = 3pq$$

$$\Rightarrow p^3 + q^2 + q = 3pq$$

- 114. (a)** Consider the line,

$$2x + 6y = 21 \quad \dots(i)$$

The point (0, 0) does not satisfy Eq. (i) but the point (0, 0) satisfy the equation $2x + 6y \leq 21$.

Now, consider the line,

$$5x - 2y = 10 \quad \dots(ii)$$

The point (0, 0) does not satisfy

Eq. (ii) but the point (0, 0) satisfy the equation $5x - 2y \leq 10$.

- 115. (b)** We have, $A(x) = x^2 + px + q$

$\therefore (x-m)$ and $(x-km)$ are factors of $A(x)$, then m and km are roots of $A(x) = 0$

$$\therefore m + km = -p \Rightarrow m(k+1) = -p$$

$$\Rightarrow m = \frac{-p}{(k+1)} \quad \dots(i)$$

$$\text{and } m \cdot km = q \Rightarrow m^2k = q$$

$$\Rightarrow \frac{p^2}{(k+1)^2} \cdot k = q \quad [\text{from Eq. (i)}]$$

$$\Rightarrow (k+1)^2 q = kp^2$$

- 116. (c)** Given points are $P(5, -1), Q(3, -2)$ and $R(1, 1)$.

If points P, Q and R lie in the solution of inequations $x + y \leq 4$ and $x - y \geq 2$, then points P, Q and R satisfy these inequations.

$$\text{For point } P(5, -1), \quad 5 - 1 \leq 4 \quad [\text{true}]$$

$$5 + 1 \geq 2 \quad [\text{true}]$$

$$\text{For point } Q(3, -2), \quad 3 - 2 \leq 4 \quad [\text{true}]$$

$$3 + 2 \geq 2 \quad [\text{true}]$$

$$\text{and for point } R(1, 1), \quad 1 + 1 \leq 4 \quad [\text{true}]$$

$$1 - 1 \geq 2 \quad [\text{false}]$$

So, the point $R(1, 1)$ does not satisfy the inequations.

Hence, points P and Q lie in the solution of inequations.

- 117. (a)** Given, $x = \frac{\sqrt{a+2b} + \sqrt{a-2b}}{\sqrt{a+2b} - \sqrt{a-2b}}$

By rationalising, we get

$$x = \frac{(\sqrt{a+2b})^2 + (\sqrt{a-2b})^2 + 2\sqrt{(a+2b)(a-2b)}}{(\sqrt{a+2b})^2 - (\sqrt{a-2b})^2}$$

$$\Rightarrow x = \frac{a+2b+a-2b+2\sqrt{a^2-4b^2}}{a+2b-a-2b}$$

$$\Rightarrow x = \frac{2a+2\sqrt{a^2-4b^2}}{4b}$$

$$\Rightarrow x = \frac{a+\sqrt{a^2-4b^2}}{2b}$$

$$\Rightarrow bx^2 - ax + b = 0$$

- 118. (a)** Given, $\sqrt{3x^2 - 7x - 30}$

$$= (x-5) + \sqrt{2x^2 - 7x - 5}$$

On squaring both sides, we get

$$3x^2 - 7x - 30 = (x-5)^2 + (2x^2 - 7x - 5)$$

$$+ 2(x-5)\sqrt{2x^2 - 7x - 5}$$

$$\Rightarrow 3x^2 - 7x - 30 = x^2 + 25 - 10x$$

$$+ 2x^2 - 7x - 5$$

$$+ (2x-10)\sqrt{2x^2 - 7x - 5}$$

$$\Rightarrow 3x^2 - 7x - 30 - x^2 + 10x$$

$$+ 7x - 2x^2 - 20$$

$$= (2x-10)\sqrt{2x^2 - 7x - 5}$$

$$\Rightarrow (10x-50)$$

$$= (2x-10)\sqrt{2x^2 - 7x - 5}$$

$$\Rightarrow 10(x-5) = 2(x-5)\sqrt{2x^2 - 7x - 5}$$

$$\Rightarrow \sqrt{2x^2 - 7x - 5} = 5 \quad \dots(i)$$

Again, on squaring both sides of Eq. (i), we get

$$2x^2 - 7x - 5 = 25$$

$$\Rightarrow 2x^2 - 7x - 30 = 0 \quad \dots(ii)$$

If α and β are roots of Eq. (ii), then

$$\alpha\beta = \frac{-30}{2} = -15$$

- 119. (a)** Since, the roots of equation

$$lx^2 + mx + m = 0$$

are in the ratio $p:q$.

Then, we can take roots as pk and qk .

$$\therefore \text{Sum of roots} = pk + qk = -\frac{m}{l}$$

$$\Rightarrow (p+q)k = -\frac{m}{l} \quad \dots(i)$$

and product of roots

$$= (pq)k^2 = \frac{m}{l} \quad \dots(ii)$$

On dividing Eq. (i) by Eq. (ii), we get

$$\frac{(p+q)k}{(pq)k^2} = -1$$

$$\Rightarrow \left(\frac{p+q}{pq}\right) = -k = \frac{m/l}{(p+q)}$$

[using Eq. (i)]

$$\Rightarrow \frac{(p+q)^2}{pq} = \frac{m}{l} \Rightarrow \frac{p+q}{\sqrt{pq}} = \pm \sqrt{\frac{m}{l}}$$

$$\Rightarrow \frac{p+q}{\sqrt{pq}} = -\sqrt{\frac{m}{l}}$$

$$\Rightarrow \sqrt{\frac{p}{q}} + \sqrt{\frac{q}{p}} + \sqrt{\frac{m}{l}} = 0$$

- 120. (d)** We have, $1 + \frac{1}{x} - \frac{1}{x^2} \geq 0$

$$\Rightarrow \frac{x^2 + x - 1}{x^2} \geq 0$$

$$\text{As, } x^2 \geq 0, \text{ then } x^2 + x - 1 \geq 0$$

$$\text{If } x^2 + x - 1 = 0,$$

$$\text{then } x = \frac{-1 \pm \sqrt{5}}{2}$$

$$\therefore \left[x - \left(\frac{-1+\sqrt{5}}{2} \right) \right] \left[x - \left(\frac{-1-\sqrt{5}}{2} \right) \right] \geq 0$$

$$\Rightarrow x \leq \frac{-1-\sqrt{5}}{2} \text{ or } x \geq \frac{-1+\sqrt{5}}{2}$$

19

SET THEORY

Usually (1-2) questions have been asked from this chapter. Generally questions are asked from these topics more results on operations on set.



SET

A well defined collection of objects, is called a set. The objects in a set are called its members or elements. Sets are usually denoted by the capital letters A, B, C, X, Y and Z etc. And the elements of a set are denoted by small letters a, b, c etc. If x is an element of set A , we can write $x \in A$, which means that ' x belongs to A ' or that x is an element of A .

If x does not belong to A , we can write, $x \notin A$. e.g.

- The collection of vowels in English alphabet is a set A containing five elements namely a, e, i, o and u , where $a \in A$ but $b \notin A$.
- The collection of first four prime numbers is a set A containing the elements 2, 3, 5 and 7, where $3 \in A$ but $1 \notin A$.

Representation of Sets

Sets are generally represented by following two ways:

1. **Roster or Tabular form or listing method** In this method, all the elements of a set are listed, within curly braces $\{ \}$ being separated by commas. e.g.
 - (i) If A is a set of first eight prime numbers, then $A = \{2, 3, 5, 7, 11, 13, 17, 19\}$
 - (ii) If B is a set of squares of first five natural numbers, then $B = \{1, 4, 9, 16, 25\}$
 - (iii) If A is a set of vowels of English alphabets, then $A = \{a, e, i, o, u\}$

Note The order in which the elements are written in a set makes no difference and also the repeated elements are taken only once each.

2. **Set Builder form or Rule method** In this method, instead of listing all the elements of a set, we write the set by some special property or properties satisfied by all its elements and write it as.
 - (i) The set B of all even natural numbers can be written as
 $B = \{x : x \text{ is a natural number and } x = 2n \text{ for } n \in N\}$ or $B = \{x : x \in N, x = 2n, n \in N\}$ and reads it as ' B ' is the set consisting of all elements x such that x has the property of even natural number. The symbol ':' or '1' stands for such that
 - (ii) The set $A = \{3, 5, 7, 9, 11\}$, then it is represented as
 $A = \{x : x = 2n + 1 \text{ where } n \in N, n < 6\}$

- (iii) The set $A = \{0, 1, 4, 9, 16, \dots\}$ can be written as $A = \{x : x = n^2, n \in \mathbb{Z}\}$, where $\mathbb{Z}$ is the set of integers.

Types of Sets

- Empty set** A set which does not contain any element is called an empty set or null set or void set. It is denoted by ϕ or $\{\}$.
e.g. A = set of all odd numbers divisible by 2
and $B = \{x : x \in \mathbb{N} \text{ and } 5 < x < 6\}$
The sets which have atleast one element are called non void or non-empty set.
- Singleton set** A set consisting of a single element is called a singleton set. e.g.
(i) The set $\{5\}$ is a singleton set.
(ii) $\{x | x \in \mathbb{W} \text{ and } x + 6 = 6\} = \{0\}$
which is a singleton set.
- Finite set** A set which consists of a definite number of elements, is called a finite set. Empty set is also a finite set. e.g.
(i) The set $\{1, 2, 3, 4\}$ is a finite set, because it contains a definite number of elements i.e. only 4 elements.
(ii) B = Set of vowels in English Alphabets
 $= \{a, e, i, o, u\}$;
(iii) A = Set of even prime natural numbers.
 $\Rightarrow A = \{2\}$

Note The number of distinct elements contained in a set A is called the cardinal number of A and is denoted as $n(A)$.
If $A = \{a, e, i, o, u\}$, then $n(A) = 5$.

- Infinite set** A set which consists of infinite number of elements is called an infinite set. It is represented by writing a few elements of the set followed by
e.g.
(i) Set of square of natural numbers is an infinite set, because natural numbers are infinite and it can be represented as $\{1, 4, 9, 16, 25, \dots\}$.
(ii) Set of all points in a plane.
- Equal sets** Two sets A and B are said to be equal, if they have exactly the same elements and we write $A = B$. Otherwise, two sets are said to be unequal and we write $A \neq B$.
e.g. Let $A = \{1, 2, 3, 4\}$ and $B = \{4, 3, 1, 2\}$, then $A = B$ because each element of A is in B and *vice-versa*.
- Equivalent sets** Two sets A and B are equivalent, if their cardinal numbers are same i.e. $n(A) = n(B)$ and we write $A \leftrightarrow B$ or $A \sim B$
e.g. Let $A = \{1, 2, 3\}$ and $B = \{4, 5, 6\}$,
then $n(A) = n(B)$
 $A \leftrightarrow B$ or $A \sim B$

- Subsets** Let A and B be two sets. If every element of A is an element of B , then A is called a subset of B .
If A is subset of B , then we can write $A \subseteq B$ which is read as ' A is a subset of B ' or ' A is contained in B '.
(i) Every set is a subset of itself i.e. $A \subseteq A, B \subseteq B$.
(ii) Empty set is a subset of every set i.e. $\phi \subseteq A$.
e.g. Let $A = \{2, 4, 6\}$ and $B = \{6, 4, 2, 8\}$.
Then, $A \subseteq B$ but $B \not\subseteq A$ i.e. A is a subset of B but B is not a subset of A .

Note The total number of subsets of a finite set containing n elements is 2^n .

- Super set** If A is a subset of B , then we say that B is superset of A and we write $B \supseteq A$.
e.g. If $A = \{1, 2, 3, 4\}$ and $B = \{1, 2, 3, 4, 5, 6\}$,
then, $B \supseteq A$.
- Comparability of sets** Two sets A and B are said to be comparable, if either $A \subset B$ or $B \subset A$ or $A = B$, otherwise A and B are said to be incomparable.
e.g. Let $A = \{1, 2, 3\}$, $B = \{1, 2, 4, 6\}$ and $C = \{1, 2, 4\}$
Since, $A \not\subseteq B$ and $B \not\subseteq A$.
So, A and B are incomparable but $C \subset B$ and so B and C are comparable sets.
- Proper subset** If $A \subseteq B$ and $A \neq B$, then A is called a proper subset of B and we write $A \subset B$.
e.g. If $A = \{1, 2, 3, 4, \dots\}$
and $B = \{\dots, -3, -2, -1, 0, 1, 2, 3, \dots\}$,
then $A \subset B$.

Note The total number of proper subset of a finite set containing n elements is $(2^n - 1)$.

- Universal set** If there are some sets under consideration, then there happens to be a set which is a superset of each one of the given sets. Such a set is known as the universal set and it is denoted by U .
e.g.
• If $A = \{1, 2, 3\}$, $B = \{2, 3, 4, 5, 7\}$ and $C = \{2, 4, 6, 8\}$, then $U = \{1, 2, 3, 4, 5, 6, 7, 8\}$ is a universal set for A, B and C .
• For the set of all integers, the universal set can be the set of rational numbers or the set of real numbers.
- Power set** The collection of all subsets of a set A , is called power set of A and it is denoted by $P(A)$. In $P(A)$, every element is a set.
e.g. Let $A = \{1, 2, 3\}$
Then, $P(A) = \{\phi, \{1\}, \{2\}, \{3\}, \{1, 2\}, \{2, 3\}, \{3, 1\}, \{1, 2, 3\}\}$

Properties of Power Sets

1. Each element of a power set is a set.
2. If set A has n elements, then $P(A)$ has 2^n elements.
3. If A is an empty set ϕ or $\{\}$, then $P(A)$ has just one element i.e. $P(A) = \{\phi\}$.
4. If $A \subseteq B$, then $P(A) \subseteq P(B)$.

EXAMPLE 1. The set $S = \{x \in N : x + 3 = 3\}$ is a

- a. null set b. singleton set
c. infinite set d. None of these

Sol. a. Given, $S = \{x \in N : x + 3 = 3\}$

$\therefore S = \{\}$
So, S is a null set.

EXAMPLE 2. If $A = \{x : x \text{ is an odd integer}\}$ and $B = \{x : x^2 - 8x + 15 = 0\}$. Then, which one of the following is correct.

☑ 2013 I

- a. $A = B$ b. $A \subseteq B$ c. $B \subseteq A$ d. $A \subseteq B^c$

Sol. c. Given that, $A = \{x : x \text{ is an odd integer}\}$

and $B = \{x : x^2 - 8x + 15 = 0\} = \{x : x^2 - 5x - 3x + 15 = 0\}$
 $= \{x : x(x - 5) - 3(x - 5) = 0\}$
 $= \{x : (x - 5)(x - 3) = 0\} = \{3, 5\}$

Since, B has two odd elements.
 $\therefore B \subseteq A$

EXAMPLE 3. If $A = \{a, b\}$, then power set of A is

- a. $\{\phi, a, b\}$ b. $\{\phi, a, b, A\}$
c. $\{\phi, a, b, ab\}$ d. $\{\phi, \{a\}, \{b\}, A\}$

Sol. d. $A = \{a, b\}$

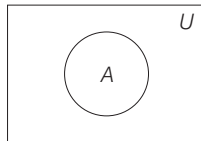
The subsets of A are $\phi, \{a\}, \{b\}, \{a, b\}$.

$\therefore P(A) = \{\phi, \{a\}, \{b\}, \{a, b\}\} = \{\phi, \{a\}, \{b\}, A\}$

VENN DIAGRAM

‘Venn diagram’ is the diagrammatic representation of various types of sets and operations on sets.

The universal set is usually represented by a rectangular region and other subsets of the universal set are represented by circles inscribed in it. Each inscribed circle represents a set (subset of universal set).



Operations on Sets

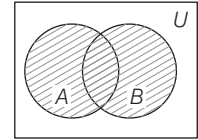
1. **Union of Sets** Let A and B be any two sets. The union of A and B is the set of all the elements of A and all the elements of B , the common elements being taken only once. It is denoted by $A \cup B$ and read as ‘ A union B .’

Thus, $A \cup B = \{x : x \in A \text{ or } x \in B\}$

In the given Venn diagram $A \cup B$ is denoted by the shaded region.

e.g. If $A = \{1, 2, 3, 4\}$

and $B = \{1, 2, 3, 5, 7\}$, then $A \cup B = \{1, 2, 3, 4, 5, 7\}$



Note • $x \in A \cup B \Leftrightarrow x \in A \text{ or } x \in B$

• $x \notin A \cup B \Leftrightarrow x \notin A \text{ and } x \notin B$ • $A \subseteq A \cup B, B \subseteq A \cup B$

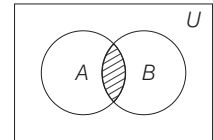
2. **Intersection of sets** Let A and B be any two sets. The intersection of A and B is the set of all those elements which belongs to both A and B . It is denoted by $A \cap B$ and read as A intersection B .

Thus, $A \cap B = \{x : x \in A \text{ and } x \in B\}$

$A \cap B$ is represented by the shaded region in the given Venn diagram

e.g. If $A = \{1, 2, 3, 4\}$ and

$B = \{1, 3, 7, 9\}$, then $A \cap B = \{1, 3\}$.



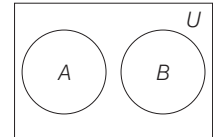
Note • $x \in A \cap B \Leftrightarrow x \in A \text{ and } x \in B$

• $x \notin A \cap B \Leftrightarrow x \notin A \text{ or } x \notin B$ • $A \cap B \subseteq A$ and $A \cap B \subseteq B$

3. **Disjoint sets** Two sets A and B are said to be disjoint, if they have no common element i.e. $A \cap B = \phi$

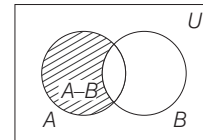
e.g. If $A = \{1, 2, 4\}$, $B = \{3, 5, 6\}$, then $A \cap B = \phi$

So, A and B are disjoint sets.



4. **Difference of sets** Let A and B be two sets. The difference of two sets is the set of all those elements of A which do not belong to B . It is denoted by $A - B$ and read as A minus B .

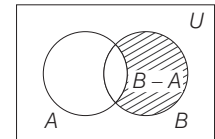
$\therefore A - B = \{x : x \in A \text{ and } x \notin B\}$



The shaded part in the given Venn diagram represents $A - B$.

Similarly, the difference $B - A$ is the set of all those elements of B that do not belong to A .

$\therefore B - A = \{x : x \in B \text{ and } x \notin A\}$



The shaded part in the given Venn diagram represents $B - A$.

e.g.

- If $A = \{2, 4, 6, 8, 10\}$ and $B = \{2, 4, 6, 12, 14\}$, then $A - B = \{8, 10\}$ and $B - A = \{12, 14\}$.
- If $A = \{1, 2, 3\}$ and $B = \{1, 2, 3, 4\}$, then $A - B = \phi$.